Olympus, after the home of the Greek gods—is more than 25 kilometers (roughly 15 miles) high, dwarfing not only the largest volcano on Earth but also the largest mountain of any sort, Mt. Everest, which stands 9 kilometers above the Tibetan plateau. There are some 20 large volcanos on Mars, but none so massive as Olympus Mons, which has a volume about 100 times that of the largest volcano on Earth, Mauna Loa in Hawaii.

By counting the accumulated impact craters (made by small impacting asteroids, and readily distinguished from summit calderas) on the flanks of the volcanos, estimates of their ages can be derived. Some Martian volcanos turn out to be a few billion years old, although none dates back to the very origin of Mars, about 4.5 billion years ago. Some, including Olympus Mons, are comparatively new—perhaps only a few hundred million years old. It is clear that enormous volcanic explosions occurred early in Martian history, perhaps providing an atmosphere much denser than the one Mars holds today. What would the place have looked like if we had visited it then?

Some volcanic flows on Mars (for example, in Cerberus) formed as recently as 200 million years ago. It is, I suppose, even possible—although there is no evidence either way—that Olympus Mons, the largest volcano we know about for certain in the Solar System, will be active again. Volcanologists, a patient sort, would doubtless welcome the event.

In 1990–93 the *Magellan* spacecraft returned surprising radar data about the landforms of Venus. Cartographers prepared maps of almost the entire planet, with fine detail down to about 100 meters, the goal-line-to-goal-line distance in an American football stadium. More data were radioed home by *Magellan* than by all other planetary missions combined. Since much of the ocean floor remains unexplored (except perhaps for still-classified data acquired by the U.S. and Soviet navies), we may know more about the surface topography of Venus than about any other planet, Earth included. Much of the geology of Venus is unlike anything seen on Earth or anywhere else. Planetary geologists have given these landforms